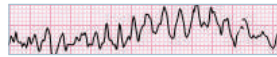


CARDIOPULMONARY RESUSCITATION (CPR)

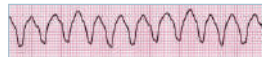
In the patient with no pulse, record the time and start chest compressions:

- Give continuous cycles of 30 chest compressions and 2 breaths with bag-valve-mask attached to 100% oxygen at 10-15L/min.
- Attach monitor/defibrillator and pause compressions to check initial heart rhythm:

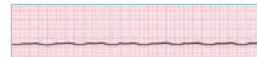
Ventricular fibrillation (VF)



Pulseless ventricular tachycardia (pVT)



Asystole



**Any other rhythm:
Pulseless electrical activity (PEA)**



Give shock of 120-150J.
If monophasic defibrillator, give instead shock of 360J.

- Immediately restart CPR, starting with compressions.
- After 2 minutes of CPR, pause compressions and recheck heart rhythm:

VF

pVT

Asystole

Other rhythm

Give shock of 120-150J (increase joules with each shock given).
If monophasic defibrillator, give instead shock of 360J.

Feel for carotid pulse for up to 10 seconds.

No pulse felt

Unsure

Pulse felt

PEA

Stop CPR
and check
breathing
→ 14.

- Immediately restart CPR, starting with compressions.
- Give **adrenaline**¹ 1mL (1:1000 solution) IV, followed by 5mL **sterile water** or **sodium chloride 0.9%**. Repeat adrenaline every 2 cycles (every 3-5 minutes).
- After every 2 minutes of CPR, pause compressions, recheck heart rhythm and manage as above.

While giving continuous CPR:

- If **VF** or **pVT**: after 3rd shock, give **amiodarone** 300mg IV, followed by 10mL **sodium chloride 0.9%**.
- If VF or pVT persists after next shock or recurs, give further **amiodarone** 150mg IV.
- Doctor to consider intubation. If intubated, give 1 breath every 6 seconds and continuous chest compressions.
- Look for and manage possible cause:
- If trauma, diarrhoea/vomiting or dehydration, give **sodium chloride 0.9%** 1L IV rapidly. Repeat if needed. If unsure, discuss with doctor.
- If glucose < 3 or unable to measure > 17. If temperature ≤ 35°C > 16. If overdose/poisoning, discuss with specialist or local poison helpline > 178.
- If more resonant/decreased breath sounds on 1 side or deviated trachea, **tension pneumothorax** likely: insert large bore cannula above 3rd rib in mid-clavicular line.

Decide when to stop CPR:

- If no pulse after 30 minutes of continuous CPR:
- If ongoing VF/pVT, temperature ≤ 35°C or overdose/poisoning, continue CPR and discuss/transfer urgently.
- If none of above, stop CPR and pronounce dead. Arrange bereavement counselling for family.

How to give chest compressions

- Ensure patient is lying on firm surface. If on bed, use backboard or move patient onto floor.
- Place heel of one hand over lower half of sternum. Place heel of second hand on top of first hand.
- Push down quickly, hard (depth of 5-6cm) and fast (100-120 per minute).
- Allow chest to return to normal shape between compressions.
- Do not interrupt compressions unless giving ventilations or checking heart rhythm.
- Swap with colleague every 2 minutes to avoid fatigue.

¹Adrenaline is also known as epinephrine.